

- ♦ unusual acoustic behavior caused by exhaust flaps and acoustic simulators

3. Lack of brake boost.

Diagnostic function

- ♦ The vacuum sensor is used to check whether the pressure in the system has fallen below a certain value in a certain time.
- ♦ The **System test: vacuum system** in the PIWIS Tester can be used for complete testing of the vacuum system because all solenoid switchover valves are forcibly actuated at the time of the diagnostic test in this case.
 - i Very small leaks might not be detected during the system test.

i INFORMATION

1. The vacuum system includes:

- ♦ The mechanical vacuum pump
- ♦ The brake booster circuit with:
 - ♦ Brake booster
 - ♦ Line with check valve to brake booster
 - ♦ Brake vacuum sensor between check valve and brake booster
- ♦ The auxiliary load circuit with:
 - ♦ Coolant shutoff valve and associated solenoid switchover valve, respectively for:
 - ♦ Heater
 - ♦ Engine
 - ♦ Gearwheel oil (transmission)
 - ♦ Clutch fluid (PDK only)
 - ♦ Vacuum unit and associated solenoid switchover valve, respectively for:
 - ♦ Switched coolant pump
 - ♦ Diverter valves
 - ♦ Boost pressure control valves (wastegates)
 - ♦ Exhaust flaps
 - ♦ Acoustic simulators
- ♦ Vacuum lines between the branch for auxiliary electrical loads and all aforementioned components.

2. There may be several small leaks, which correspond in total to the leak determined here.

Diagnostic information - DME (DFI) control unit

Vacuum system leaking - cooling system fault

Diagnostic conditions

- Perform system test on vacuum system

OR complete the following two steps one after the other:

1. Establish the following conditions three times one after the other:

- Vehicle is stationary:
 - Engine oil temperature (T030) 30 to 110 °C (86 to 230 °F)
 - Engine idling
 - Forcefully apply the brake several times (brake pressure (A255) greater than 12 bar) until the vacuum (A250) increases to over 500 hPa
 - Then do not actuate the brake for more than 40 seconds
 - Then drive a short distance at a speed of over 2 km/h (1 mph), maintaining an engine speed of over 1,500 rpm

2. Then establish the following conditions three times one after the other:

- Actuate the brake at least once while driving (brake pressure (A255) greater than 5.5 bar) - the vacuum (A250) must increase to over 200 hPa
- Then drive at as constant an engine speed (A020) as possible between 1,000 ... 6,000 rpm for more than 60 seconds

❗ If vacuum-actuated valves are switched during the 60 seconds of constant driving, the diagnostic process will be interrupted and will start from the very beginning again after the gear change so that the time that must be driven in order to complete diagnostics can be much longer!

Possible fault causes

- ♦ Vacuum system leaking (see ⓘ **INFORMATION**)
- ♦ Brake light switch faulty
- ♦ Brake light switch aligned incorrectly at brake pedal
- ♦ Contact resistance on plug connection to the brake vacuum sensor
- ♦ Brake vacuum sensor faulty

Fault effects

1. The message [\[Cooling system fault\]](#) appears in the instrument cluster if a leak is detected in the vacuum system.

Reason: The cooling system with thermal management also includes vacuum-actuated shutoff valves.

2. If there is no vacuum, the vacuum-actuated flaps and valves can no longer be closed, which can lead to malfunctions.

Noticeable effects can include:

- ♦ lack of torque
- ♦ no heating power

- ♦ unusual acoustic behavior caused by exhaust flaps and acoustic simulators

3. Lack of brake boost.

Diagnostic function

- ♦ The vacuum sensor is used to check whether the pressure in the system has fallen below a certain value in a certain time.
- ♦ The **System test: vacuum system** in the PIWIS Tester can be used for complete testing of the vacuum system because all solenoid switchover valves are forcibly actuated at the time of the diagnostic test in this case.
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1. The vacuum system includes:

- ♦ The mechanical vacuum pump
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 - ♦ Brake booster
 - ♦ Line with check valve to brake booster
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- ♦ The auxiliary load circuit with:
 - ♦ Coolant shutoff valve and associated solenoid switchover valve, respectively for:
 - ♦ Heater
 - ♦ Engine
 - ♦ Gearwheel oil (transmission)
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 - ♦ Vacuum unit and associated solenoid switchover valve, respectively for:
 - ♦ Switched coolant pump
 - ♦ Diverter valves
 - ♦ Boost pressure control valves (wastegates)
 - ♦ Exhaust flaps
 - ♦ Acoustic simulators
- ♦ Vacuum lines between the branch for auxiliary electrical loads and all aforementioned components.

2. There may be several small leaks, which correspond in total to the leak determined here.

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 1: overtemperature

- ♦ 911 Carrera - right
- ♦ 718 Boxster/Cayman - left

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

- ♦ **ALWAYS** keep hair or clothing out of the rotating range of the fan.
- ♦ DO NOT reach into the rotating fan.
- ♦ **DO NOT** place tools or other items in the rotating range of the fan.
- ♦ Perform work on the fan only with the ignition switched off.
- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

i INFORMATION

- ♦ The engine compartment purge fan is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the engine compartment purge fan can report faults to the DME control unit.

Diagnostic conditions

- Engine compartment purge fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Air intake for engine compartment purge fan 1 covered (car cover)
- ♦ Engine compartment purge fan sluggish
- ♦ Engine compartment purge fan 1 faulty

Fault setting condition(s)

- Overtemperature in engine compartment purge fan 1

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 1: mechanical fault

- ♦ 911 Carrera - right
- ♦ 718 Boxster/Cayman - left

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Diagnostic conditions

- Engine compartment purge fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Engine compartment purge fan 1 sluggish or blocked
- ♦ Engine compartment purge fan 1 faulty

Fault setting condition(s)

- Current draw of engine compartment purge fan 1 too high

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 1: electrical fault

- ♦ 911 Carrera - right
- ♦ 718 Boxster/Cayman - left

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Diagnostic conditions

- Engine compartment purge fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Engine compartment purge fan 1 faulty

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 1: internal power supply

- ♦ 911 Carrera - right
- ♦ 718 Boxster/Cayman - left

⚠ WARNING

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- ♦ The engine compartment purge fan is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the engine compartment purge fan can report faults to the DME control unit.

Diagnostic conditions

- Engine compartment purge fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Short circuit to ground or open circuit in lines (power supply to engine compartment purge fan terminal 30)
- ♦ Engine compartment purge fan 1 faulty

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 2: overtemperature

- ♦ 911 Carrera - left
- ♦ 718 Boxster/Cayman - right

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

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i INFORMATION

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- ♦ In the opposite direction, the engine compartment purge fan can report faults to the DME control unit.

Diagnostic conditions

- Engine compartment purge fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Air intake for engine compartment purge fan 2 covered (car cover)
- ♦ Engine compartment purge fan sluggish
- ♦ Engine compartment purge fan 2 faulty

Fault setting condition(s)

- Overtemperature in engine compartment purge fan 2

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 2: mechanical fault

- ♦ 911 Carrera - left
- ♦ 718 Boxster/Cayman - right

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Exposed running fan blades!

Danger of limbs being severed!

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i INFORMATION

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- ♦ In the opposite direction, the engine compartment purge fan can report faults to the DME control unit.

Diagnostic conditions

- Engine compartment purge fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Engine compartment purge fan 2 sluggish or blocked
- ♦ Engine compartment purge fan 2 faulty

Fault setting condition(s)

- Current draw of engine compartment purge fan 2 too high

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 2: electrical fault

- ♦ 911 Carrera - left
- ♦ 718 Boxster/Cayman - right

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

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i INFORMATION

- ♦ The engine compartment purge fan is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the engine compartment purge fan can report faults to the DME control unit.

Diagnostic conditions

- Engine compartment purge fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Engine compartment purge fan 2 faulty

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 2: internal power supply

- ♦ 911 Carrera - left
- ♦ 718 Boxster/Cayman - right

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

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- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

NOTE

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i INFORMATION

- ♦ The engine compartment purge fan is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the engine compartment purge fan can report faults to the DME control unit.

Diagnostic conditions

- Engine compartment purge fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Short circuit to ground or open circuit in lines (power supply to engine compartment purge fan terminal 30)
- ♦ Engine compartment purge fan 2 faulty

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 1: live code implausible

- ♦ 911 Carrera - right
- ♦ 718 Boxster/Cayman - left

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

- ♦ **ALWAYS** keep hair or clothing out of the rotating range of the fan.
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- ♦ Perform work on the fan only with the ignition switched off.
- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

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Sensitive electronics!

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- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in power supply for engine compartment purge fan 1
- ♦ Open circuit in ground line for engine compartment purge fan 1
- ♦ Engine compartment purge fan 1 faulty

Fault setting condition(s)

The DME control unit expects to receive a ground signal cyclically on the activation line from the engine compartment purge fan. The fault is set if this signal is not received even though the activation line itself is not faulty.

Diagnostic information - DME (DFI) control unit

Engine compartment purge fan 2: live code implausible

- ♦ 911 Carrera - left
- ♦ 718 Boxster/Cayman - right

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

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- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in power supply for engine compartment purge fan 2
- ♦ Open circuit in ground line for engine compartment purge fan 2
- ♦ Engine compartment purge fan 2 faulty

Fault setting condition(s)

The DME control unit expects to receive a ground signal cyclically on the activation line from the engine compartment purge fan. The fault is set if this signal is not received even though the activation line itself is not faulty.

Diagnostic information - DME (DFI) control unit

Relay for radiator fan 2 stuck closed

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

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- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

Diagnostic conditions

- Engine running, then:
 - Ignition off for more than 1 minute

Possible fault causes

- ♦ Short circuit to B+ in line from the relay to the fan
- ♦ Relay faulty

Fault setting condition(s)

After the ignition is switched off, the fan reports that there is still voltage present at terminal 30.

Diagnostic information - DME (DFI) control unit

Relay for radiator fan 2 stuck closed

⚠ WARNING

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Diagnostic conditions

- Engine running, then:
 - Ignition off for more than 1 minute

Possible fault causes

- ♦ Short circuit to B+ in line from the relay to the fan
- ♦ Relay faulty

Fault setting condition(s)

After the ignition is switched off, the fan reports that there is still voltage present at terminal 30.

Diagnostic information - DME (DFI) control unit

Radiator fan 1: overtemperature

⚠ WARNING

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NOTE

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- ♦ **DO NOT** carry out electrical tests.
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i INFORMATION

- ♦ The control unit is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the control unit can report faults to the DME control unit.

Diagnostic conditions

- Radiator fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Air intake for radiator 1 covered (car cover)
- ♦ Fan sluggish
- ♦ Control unit for radiator fan 1 faulty

Fault setting condition(s)

- Overtemperature in control unit for radiator fan 1

Fault effects

- ♦ Radiator fan 1 switched off

Diagnostic information - DME (DFI) control unit

Radiator fan 1: electrical fault

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Diagnostic conditions

- Radiator fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Radiator fan 1 faulty
- ♦ Control unit for radiator fan 1 faulty

Fault effects

- ♦ Radiator fan 1 switched off

Diagnostic information - DME (DFI) control unit

Radiator fan 1: internal power supply

⚠ WARNING

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- ♦ The control unit is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the control unit can report faults to the DME control unit.

Diagnostic conditions

- Radiator fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Short circuit to ground or open circuit in lines (power supply to radiator fan control unit terminal 30)
- ♦ Relay faulty
- ♦ Radiator fan 1 faulty
- ♦ Control unit for radiator fan 1 faulty

Diagnostic information - DME (DFI) control unit

Radiator fan 2: overtemperature

⚠ WARNING

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Diagnostic conditions

- Radiator fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Air intake for radiator 2 covered (car cover)
- ♦ Fan sluggish
- ♦ Control unit for radiator fan 2 faulty

Fault setting condition(s)

- Overtemperature in control unit for radiator fan 2

Fault effects

- ♦ Radiator fan 2 switched off

Diagnostic information - DME (DFI) control unit

Radiator fan 2: electrical fault

⚠ WARNING

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- ♦ The control unit is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the control unit can report faults to the DME control unit.

Diagnostic conditions

- Radiator fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Radiator fan 2 faulty
- ♦ Control unit for radiator fan 2 faulty

Fault effects

- ♦ Radiator fan 2 switched off

Diagnostic information - DME (DFI) control unit

Radiator fan 2: internal power supply

⚠ WARNING

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i INFORMATION

- ♦ The control unit is activated by the DME control unit via a signal line.
- ♦ In the opposite direction, the control unit can report faults to the DME control unit.

Diagnostic conditions

- Radiator fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Short circuit to ground or open circuit in lines (power supply to radiator fan control unit terminal 30)
- ♦ Relay faulty
- ♦ Radiator fan 2 faulty
- ♦ Control unit for radiator fan 2 faulty

Diagnostic information - DME (DFI) control unit

Radiator fan 1: mechanical fault

⚠ WARNING

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Diagnostic conditions

- Radiator fan 1 activated > 5 minute(s)

Possible fault causes

- ♦ Radiator fan 1 sluggish or blocked
- ♦ Radiator fan 1 faulty

Fault setting condition(s)

- Current draw for radiator fan 1 too high

Fault effects

- Sluggish:
 - ♦ Radiator fan 1 speed reduced to 0 rpm
- Blocked:
 - ♦ Engine shakes loose

Diagnostic information - DME (DFI) control unit

Radiator fan 2: mechanical fault

⚠ WARNING

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Diagnostic conditions

- Radiator fan 2 activated > 5 minute(s)

Possible fault causes

- ♦ Radiator fan 2 sluggish or blocked
- ♦ Radiator fan 2 faulty

Fault setting condition(s)

- Current draw for radiator fan 2 too high

Fault effects

- Sluggish:
 - ♦ Radiator fan 2 speed reduced to 0 rpm
- Blocked:
 - ♦ Engine shakes loose

Diagnostic information - DME (DFI) control unit

Radiator fan 1: live code implausible

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- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in power supply to terminal 87 for the fan
- ♦ Open circuit in ground line for the fan
- ♦ Control unit in the fan faulty

Fault setting condition(s)

The DME control unit expects to receive a ground signal cyclically on the activation line from the fan. The fault is set if this signal is not received even though the activation line itself is not faulty.

Diagnostic information - DME (DFI) control unit

Radiator fan 2: live code implausible

⚠ WARNING

Exposed running fan blades!

Danger of limbs being severed!

Risk of serious injury due to ejected foreign bodies!

- ♦ **ALWAYS** keep hair or clothing out of the rotating range of the fan.
- ♦ DO NOT reach into the rotating fan.
- ♦ **DO NOT** place tools or other items in the rotating range of the fan.
- ♦ Perform work on the fan only with the ignition switched off.
- ♦ Secure the ignition to prevent it from being switched on accidentally by other persons.

NOTE

Sensitive electronics!

Risk of irreparable damage to electronic components!

- ♦ **DO NOT** carry out electrical tests.
- ♦ **DO NOT** connect external power sources.

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Short circuit to ground or open circuit in power supply to terminal 87 for the fan
- ♦ Open circuit in ground line for the fan
- ♦ Control unit in the fan faulty

Fault setting condition(s)

The DME control unit expects to receive a ground signal cyclically on the activation line from the fan. The fault is set if this signal is not received even though the activation line itself is not faulty.

P1477

Diagnostic information - DME (DFI) control unit

Engine coolant shutoff valve stuck closed
or
Thermostat stuck closed

Diagnostic conditions

- Uniform driving (only slight changes in rpm and torque)
- Setpoint engine temperature (T005) > 104.25 °C (219.65 °F)
- **Engine coolant temperature** (T020) = setpoint engine temperature (T005)
- **Coolant temperature** at radiator outlet (T025) < 80 °C (176 °F)
- **Engine speed** 1,200 ... 3,500 rpm
- Air conditioner OFF (= heating valve must not switch during diagnostics)

Possible fault causes

- ♦ Solenoid switchover valve for coolant shutoff valve vacuum unit mechanically stuck
- ♦ Vacuum lines on the solenoid switchover valve interchanged
- ♦ Vacuum lines or electrical lines of the solenoid switchover valve interchanged with another solenoid switchover valve
- ♦ Engine coolant shutoff valve faulty (stuck closed)
- ♦ Thermostat faulty (stuck closed)
- ♦ Air trapped in the cooling system (e.g. after repairs were performed on the cooling system)
- ♦ Incorrect measurement by engine temperature sensor or radiator return temperature sensor, but still in the plausible range (-40...+150 °C/-40...+302 °F)

Diagnostic function ⇒ **See function description**

P1478

Diagnostic information - DME (DFI) control unit

Engine coolant shutoff valve stuck closed
or
Thermostat stuck closed

Diagnostic conditions

- Uniform driving (only slight changes in rpm and torque)
- Setpoint engine temperature (T005) > 104.25 °C (219.65 °F)
- **Engine coolant temperature** (T020) = setpoint engine temperature (T005)
- **Coolant temperature** at radiator outlet (T025) < 80 °C (176 °F)
- **Engine speed** 1,200 ... 3,500 rpm
- Air conditioner OFF (= heating valve must not switch during diagnostics)

Possible fault causes

- ♦ Solenoid switchover valve for coolant shutoff valve vacuum unit mechanically stuck
- ♦ Vacuum lines on the solenoid switchover valve interchanged
- ♦ Vacuum lines or electrical lines of the solenoid switchover valve interchanged with another solenoid switchover valve
- ♦ Engine coolant shutoff valve faulty (stuck closed)
- ♦ Thermostat faulty (stuck closed)
- ♦ Air trapped in the cooling system (e.g. after repairs were performed on the cooling system)
- ♦ Incorrect measurement by engine temperature sensor or radiator return temperature sensor, but still in the plausible range (-40...+150 °C/-40...+302 °F)

Diagnostic function ⇒ **See function description**

P1509

Diagnostic information - DME (DFI) control unit

Control unit function monitoring

or

- For **DTC-DFCC** 00F068:
Starter activation with powertrain closed

 The value **DTC-DFCC** is shown in the extended fault memory for the respective fault code.

Diagnostic conditions

- Switch on ignition for more than 10 second(s)
- Then idling for more than 30 second(s)
- Then engine speed of over 1,000 rpm for more than 60 seconds with vehicle stationary
- Then drive with restart by Start Stop system

Possible fault causes

- For **DTC-DFCC** 00F056, 00F057, 00F081:
 - ♦ Power supply or ground supply faulty ⇒ See P1559
 - ♦ DME (DFI) control unit software error ► Re-program

or

- For **DTC-DFCC** 00F068:
 - ♦ DME (DFI) control unit faulty

Fault setting condition(s)

- For **DTC-DFCC** 00F056, 00F057, 00F081:
 - All calculations for functions in the control unit are monitored by a higher-level unit. A fault is entered if monitoring detects implausible calculations.

or

- For **DTC-DFCC** 00F068:
 - The starter was actuated by the Start Stop system although a closed powertrain was detected (gear engaged and clutch closed).

Fault effects

- For **DTC-DFCC** 00F068:
 - ♦ Start Stop system deactivated

P1533

Diagnostic information - DME (DFI) control unit

Switch-off time fault

Diagnostic conditions

- Engine running
- Then switch off ignition
- Immediately afterwards: ignition on > 10 minute(s)

Possible fault causes

- ♦ The power supply to the DME control unit was interrupted during control unit run-on (see fault code P0610 and U3001)
- ♦ DME control unit faulty

INFORMATION

The switch-off time is calculated in the DME control unit.

- ♦ The DME control unit receives time information from the instrument cluster at regular intervals of approx. 4 seconds when the ignition is on.
- ♦ These times are compared.

Diagnostic information - DME (DFI) control unit

Oil pump activation

(oil pressure too low)

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Oil level too low
- ♦ Oil dilution
- ♦ Dirt in oil circuit
- ♦ Oil circuit faulty (increased leakage inside the engine)
- ♦ Oil pressure sensor faulty (incorrect trimming)
- ♦ Oil pressure control valve faulty
- ♦ Oil pump faulty

Fault setting condition(s)

The measured oil pressure is compared with the setpoint oil pressure.

The setpoint oil pressure depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine ...

- The fault is detected if the oil pressure is below a defined range for more than 30 seconds.

Diagnostic information - DME (DFI) control unit

Oil pump activation

(oil pressure too high)

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Oil quality
- ♦ Oil thickening
- ♦ Dirt in oil circuit
- ♦ Oil circuit faulty
- ♦ Oil pressure sensor faulty (incorrect trimming)
- ♦ Oil pressure control valve faulty
- ♦ Oil pump faulty

Fault setting condition(s)

The measured oil pressure is compared with the setpoint oil pressure.

The setpoint oil pressure depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine ...

- The fault is detected if the oil pressure is above a defined range for more than 30 seconds.

P1538

Diagnostic information - DME (DFI) control unit

Oil pump activation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Oil quality
- ♦ Oil thickening
- ♦ Dirt in oil circuit
- ♦ Oil circuit faulty
- ♦ Oil pressure sensor faulty (incorrect trimming)
- ♦ Oil pressure control valve faulty
- ♦ Oil pump faulty

Fault setting condition(s)

The oil pressure control valve is activated via a PWM signal.

The PWM signal depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine (rpm, load, oil temperature), measured oil pressure (at the oil pressure sensor) ...

- The fault is detected if the PWM signal is below a defined range for more than 8 seconds.

Diagnostic information - DME (DFI) control unit

Oil pump activation

Diagnostic conditions

- Engine running

Possible fault causes

- ♦ Dynamic driving style when oil level is near minimum (oil foaming)
- ♦ Oil level too low
- ♦ Oil dilution
- ♦ Dirt in oil circuit
- ♦ Oil circuit faulty (increased leakage inside the engine)
- ♦ Oil pressure sensor faulty (incorrect trimming)
- ♦ Oil pressure control valve faulty
- ♦ Oil pump faulty

Fault setting condition(s)

The oil pressure control valve is activated via a PWM signal.

The PWM signal depends, among other things, on the ambient conditions (temperatures, off time, etc.), operating point of the engine (rpm, load, oil temperature), measured oil pressure (at the oil pressure sensor) ...

- The fault is detected if the PWM signal is above a defined range for more than 8 seconds.

P1546

Diagnostic information - DME (DFI) control unit

Wheel speed implausible

Diagnostic conditions

- Driving

Possible fault causes

- ♦ Fault in PSM area

Fault setting condition(s)

- The wheel speed deviates from the other wheel speeds for too long.

P1547

Diagnostic information - DME (DFI) control unit

Wheel speed implausible

Diagnostic conditions

- Driving

Possible fault causes

- ♦ Fault in PSM area

Fault setting condition(s)

- The wheel speed deviates from the other wheel speeds for too long.

P1548

Diagnostic information - DME (DFI) control unit

Wheel speed implausible

Diagnostic conditions

- Driving

Possible fault causes

- ♦ Fault in PSM area

Fault setting condition(s)

- The wheel speed deviates from the other wheel speeds for too long.

P1549

Diagnostic information - DME (DFI) control unit

Wheel speed implausible

Diagnostic conditions

- Driving

Possible fault causes

- ♦ Fault in PSM area

Fault setting condition(s)

- The wheel speed deviates from the other wheel speeds for too long.

P1553

Diagnostic information - DME (DFI) control unit

Control unit function monitoring, wake-up

INFORMATION

- ♦ The entered fault code P1553 or P1555 is merely an indication that the DME control unit performed a reboot in its internal monitoring unit when starting the engine.
- ♦ This normally remains unnoticed by the driver!
- ♦ In the event of a **customer complaint** reporting that manual engine starting (**starting with the key**) was refused, was aborted or took too long, the DME control unit is faulty.
- ♦ If **no such customer complaint** exists, the **DME control unit is OK!**

Diagnostic conditions

- Vehicle voltage > 11 V
- Switch on ignition and then start the engine

Possible fault causes

- In the event of a customer complaint reporting that manual engine starting (starting with the key) was refused, was aborted or took too long, the DME control unit is faulty.
- If **no such customer complaint** exists, the **DME control unit is OK!** ► Erase fault memory.

Fault effects

- ♦ None if this is only a once-off occurrence.
- or
- ♦ Starting with the key was refused or aborted or took too long.

P1559

Diagnostic information - DME (DFI) control unit

Control unit restart during operation (warm reset)

Diagnostic conditions

- Engine start

Possible fault causes

- ♦ Vehicle electrical system voltage < 6 V during starting process (e.g. battery discharged or faulty), e.g. due to contact resistance in the lines to the DME control unit.
- ♦ Excessive difference in internal monitoring processes in the control unit, e.g. in case of reversal of the engine's direction of rotation due to inadvertent shifting into reverse gear when the vehicle is driving forward.

Fault setting condition(s)

This fault is entered if the DME control unit performs a restart during operation (soft reboot/warm reset).

P1563

Diagnostic information - DME (DFI) control unit

Cruise control coding

Diagnostic conditions

- Ignition on

Possible fault causes

- ♦ Vehicle data incorrect
(See PIWIS Tester ► Additional menu [F7] ► Maintenance of vehicle data)
- ♦ Vehicle equipment wrong
- ♦ DME (DFI) control unit coded incorrectly - The data record does not match the vehicle.
- ♦ Steering wheel electronics control unit coded incorrectly